Igor Ignashin

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Education:

- B.Sc. in Applied Mathematics and Physics, MIPT (Moscow Institute of Physics and Technology), Faculty of Applied Mathematics and Informatics.
- M.Sc. student (2nd year), MIPT, Faculty of Applied Mathematics and Informatics.
- Department of Intelligent Data Analysis.

Technical Skills:

- Languages: Python, C++, SQL
- ML/Optimization: PyTorch, JAX, RL, LLMs, convex and nonconvex optimization, game theory
- Mathematics: stochastic processes, optimization, minimax problems, probability theory
- Data/Infra: YQL, DataLens, Nirvana, Linux, Git

Work Experience:

- Visiting Research Student, MBZUAI (under Eduard Gorbunov) — Jan 2026 – Mar 2026
  - Conducted theoretical research on LMO-based optimization methods and acceleration schemes, including analysis of Muon optimizer and Lookahead-type techniques.
- Laboratory of Mathematical Optimization Methods, MIPT / ISP RAS / BRAIn Lab — 2024–Present
  - Working on a Huawei project on neural network-based schedulers, focusing on reducing the performance gap between classical and learned approaches.
  - Proved local convergence of the ALSO method for nonconvex–μ-strongly concave minimax problems in the work *"Aligning Distributionally Robust Optimization with Practical Deep Learning Needs"*.
  - Leading multiple student research projects at YSDA and MIPT AI360 on LLM inference acceleration, distillation, multi-agent RL, game-theoretic LLM training, SGD dynamics, and applied optimization, with a focus on theory, experiments, and publication-oriented research.
  - Collaborating with Andrei Leonidov's group on studying noise in SGD and developing theoretical frameworks for non-convex optimization dynamics, as well as conducting joint research on multi-agent LLM systems in game-theoretic scenarios and social dilemmas.
  - Researched equilibrium traffic assignment problems and Frank-Wolfe algorithm modifications, with two journal publications and two conference presentations. Currently developing stochastic extensions of the LMO oracle and exploring applications to game-theoretic formulations.
- Yandex, Data Analyst Intern — June 2022 – September 2022
  - Automated a daily pipeline for ticket distribution and closure analytics for B2B executors, improving transparency and monitoring efficiency.
  - Built data processing workflows in YQL and designed dashboards in DataLens for operational insights.
  - Worked with Yandex internal tools, including Nirvana, for orchestrating data workflows.

Selected Publications:
My Google Scholar

- Aligning Distributionally Robust Optimization with Practical Deep Learning Needs. Accepted to NeurIPS 2025 Workshop.
- Submitted a paper to NeurIPS 2026 on stochastic dynamics of SGD, proposing a new perspective beyond Brownian motion assumptions (*Why SGD is not Brownian Motion: A New Perspective on Stochastic Dynamics*).
- Publication in "Computer Research and Modeling" journal: Frank-Wolfe modifications for equilibrium traffic assignment.
- Preprint: Stochastic Origin Frank-Wolfe for Traffic Assignment. Presented at TFN-2025. Accepted to Journal of Mathematical Sciences, Series B.
- Preprint: Modeling skiers flows via Wardrop equilibrium in closed capacitated networks. Presented at TFN-2025. Accepted to Journal of Mathematical Sciences, Series B.

- *Conjugate Frank-Wolfe in Machine Learning.* Presented at OPTIMA conference, accepted to CCIS journal.

Summer Schools and Hackathons:

- AIRI Institute Summer School on Artificial Intelligence 2025
- Summer School “Control, Information and Optimization” in memory of B. T. Polyak 2025
- Summer School and Hackathon in Financial Mathematics, Quantathon 2024
- 5th and 6th International Summer Schools on Financial Mathematics (2024–2025)